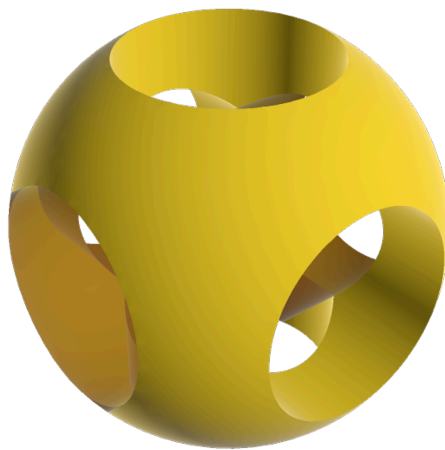




maquette-scad

Parametric CAD in Typst, via OpenSCAD + Manifold

github.com/bernsteining/maquette · bernsteining.github.io/maquette

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Introduction

maquette-scad builds 3D models right inside your Typst document. You write the geometry, it compiles, and the render lands in your PDF. There is no OpenSCAD to install and nothing to run on the side.

There are two ways to describe a model. The first is the DSL: you build the shape from Typst helpers such as `cube`, `sphere`, `difference` and `translate`, and reach for Typst's own `for`, `range` and `calc` whenever the geometry is procedural. The second is to pass the text of an existing .scad file, libraries and all.

Either way you get a watertight mesh, which the Manifold kernel guarantees, and `maquette` renders it. `render-scad` does both steps in a single call. When a shape is flat, the `-svg` variants skip the rasterizer and write 2D vector output straight out.

You can try all of this in the browser at bernsteining.github.io/maquette. Edit the SCAD, orbit the result, adjust the render, and copy the Typst it generates.

This document is the API reference: what each helper takes, what the compile calls return, and how to hand in sidecar files. Camera, lighting and post-processing belong to `maquette`, so see its manual for those. For the OpenSCAD language itself, see the OpenSCAD Users Manual.

Where to find sample .scad files

The DSL examples in this document build their geometry inline, so you need no external source to follow along. If you want to try the `compile-scad(read("..."))` path, .scad files are easy to find:

- `openscad/openscad examples/` is the set that ships with the OpenSCAD editor.
- BOSL2 is a large utility library with hundreds of documented fragments.
- Thingiverse (openscad tag) is a community archive; many entries ship the .scad alongside the .stl.

Quickstart

`render-scad` compiles and renders in one call. Pass it a DSL tree or a .scad path, followed by any of `maquette`'s render options.

```
#import "@preview/maquette-scad:0.1.0": cube, difference, render-scad, sphere

#render-scad(
  difference(
    cube(20, center: true),
    sphere(12, fn: 48),
  ),
  camera: (40, 40, 40),
  up: (0, 0, 1),
)
```

render-scad also takes source text. It is the same call, with a .scad file in place of a tree:

```
#import "@preview/maquette-scad:0.1.0": render-scad

#render-scad(read("model.scad"))
```

One call compiles and renders; no maquette import. Examples below use a show-part wrapper so the code can focus on geometry.

Config Reference

render-scad takes a few compile options, then forwards every other named argument to maquette's render-ply. All are optional; defaults are shown.

```
{ // — Compile options (maquette-scad) —————
  "fn": 32, // Default $fn segment count; a per-primitive fn: overrides it
  "smooth-normals": 30, // Crease angle (deg) for smooth vertex normals; none = faceted
  "files": {}, // Multi-file .scad parts: { "path": bytes } (or pass a read: lambda)
  "bin": {}, // Binary meshes for import(): { "name": bytes }
  "font": null // TTF/OTF bytes for text(); null = bundled DejaVu Sans
}
```

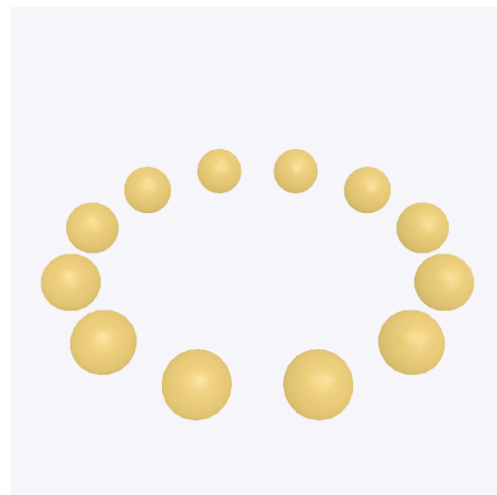
Every other argument forwards to maquette's render-ply: camera, azimuth, elevation, up, fov, zoom, background, color, shading, lights, shadows, ssao, and the rest. See the maquette manual for that full option surface. render-scad overrides two of its defaults to suit CAD facets: specular: 0 and cull_backface: false (matte and two-sided).

Building geometry with scadypst

`scadypst(tree)` walks a tree of Typst dicts and returns PLY bytes. Each helper such as `cube`, `sphere` or `translate` builds a node, and nothing runs until `scadypst` receives the whole tree.

Because the tree is Typst code, iteration and arithmetic come from Typst. Here a `..for` spread inside `union` places twelve spheres on a ring:

```
#let ring = union(..for i in range(12) {
  let a = i * 30
  let r = 20
  (translate(
    (calc.cos(a * 1deg) * r, calc.sin(a * 1deg) * r, 0),
    sphere(3, fn: 24),
  ),)
})
#show-part(scadypst(ring))
```



The variadic ops (`union`, `difference`, `hull`, `intersection`) accept `..items`, so you can spread a Typst for block to feed them a computed list.

DSL reference

Every helper takes Typst-native named arguments, as in `cube(20, center: true)`. The signatures below list positional arguments first, then named ones. `fn: N` sets the segment count for that primitive.

3D primitives

- **cube(size, center: false)**: `size` is a number for a uniform box, or a 3-array (`x`, `y`, `z`).
- **sphere(r, fn: none)**: a sphere of radius `r`.
- **cylinder(h, r: none, r1: none, r2: none, center: false, fn: none)**: pass `r` for a straight cylinder, or `r1` and `r2` for a cone or frustum.
- **polyhedron(points, faces)**: a raw mesh, where `points` is `((x, y, z), ...)` and `faces` is `((i, j, k, ...), ...)`.

2D primitives (extrusion sources)

- **square(size, center: false)**, **circle(r, fn: none)** and **ellipse(w, h, fn: none)** are the basic shapes.
- **polygon(points, paths: none)** is an arbitrary 2D shape. `paths` is a list of index-rings into `points`, where the first ring is the outer boundary and the rest are holes.
- **ngon(sides, r, fn: none)** is a regular polygon.
- **star(points, outer, inner)** is an `n`-pointed star.
- **rounded-square(w, h, r, fn: none)** is a rectangle with radius-`r` corners.
- **scad-text(str, size: 10)** draws glyph outlines from the font passed to `scadypst()`'s `font`: argument.
- **import-mesh(file)** references an STL or OBJ passed via `scadypst()`'s `bin`: dict.

2D → 3D lifting (and 3D → 2D)

- **linear-extrude(h, child, center: false, twist: 0, scale: 1, slices: none)** extrudes a 2D shape to a height `h`. `twist` is degrees over the full height, and `scale` is a taper factor from 0 to 1 or an (`sx`, `sy`) pair.
- **rotate-extrude(child, angle: 360, fn: none)** revolves a 2D profile, which lives in the `+x` half-plane, around the `z`-axis.
- **projection(child)** flattens a 3D solid to its 2D shadow on the `Z=0` plane.

Transforms (2D or 3D)

- **translate(v, child)**: `v` is a 2- or 3-vector.
- **rotate(deg, child)**: Euler degrees as (`x`, `y`, `z`), or a single number for a 2D rotation.
- **scale(v, child)**: per-axis scale.
- **mirror(v, child)**: reflect across a plane whose normal is `v`.
- **multmatrix(m, child)**: a 4×4 (or 4×3) affine matrix, the escape hatch for anything the named transforms cannot express.
- **resize(v, child)**: non-uniform scale to fit a target bounding box.
- **offset(d, child)**: grow (`d > 0`) or shrink (`d < 0`) a 2D shape by `d`.

- **color(rgb, child, alpha: none)**: rgb is a 3-array of 0..1 floats, or a 4-array (r, g, b, a). See the Colours and alpha section below.

Booleans and hulls

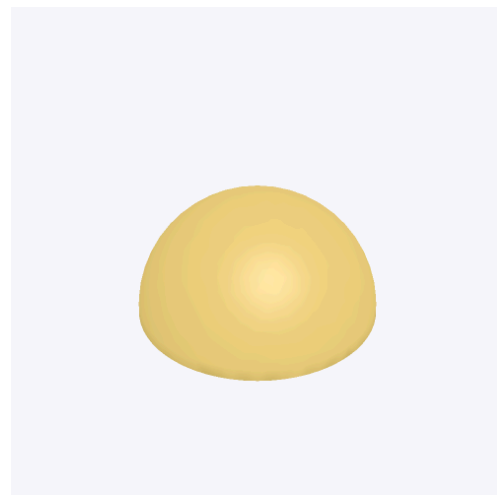
- **union(..items)**, **difference(..items)** (the first minus the rest), and **intersection(..items)**.
- **hull(..items)** is the convex hull of the union (3D).
- **hull-pts(points)** is the convex hull of a raw 3D point set, a list of (x, y, z) triples. It complements `hull()` when you have coordinates rather than geometry to wrap.
- **minkowski(..items)** is the Minkowski sum (3D).

These are all variadic, so a `..` spread over a Typst for block feeds them a computed list of children.

Plane operations

- **slice(child, z: 0)** takes a horizontal cross-section of a 3D solid at the given Z. It returns a 2D shape, which you can pipe into `scadypst-svg(..)` for a vector contour or into further 2D ops. Unlike `projection(..)`, it does not union every horizontal slice.
- **trim(child, normal, offset: 0)** cuts with a plane, keeping the half where `dot(pos, normal) ≥ offset`. It is cheaper and exact compared with the “difference() with a giant cube” trick.
- **simplify(child, epsilon: 0.01)** is a Douglas-Peucker vertex reduction that works on both 2D and 3D. Its own section has a before/after count.
- **calculate-normals(child, sharp_angle: 60)** attaches per-vertex normals for smooth shading, with a crease-angle threshold that keeps sharp edges crisp. It is 3D only, and the effect shows only under shading: “smooth”.

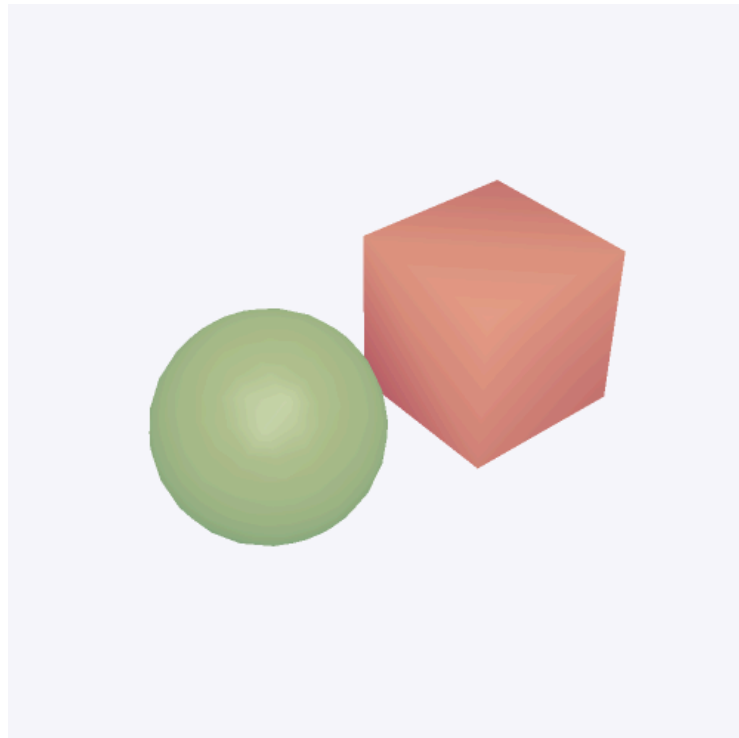
```
#show-part(scadypst(
  trim(sphere(10, fn: 64), (0, 0, 1))
))
```



Colours and alpha

Colours propagate through boolean operations. The emitted PLY carries per-vertex RGB and `render-ply` picks it up automatically. Any part without a `color()` wrapper inherits the render config’s `color: argument`.

```
#show-part(
  scadypst(union(
    color((0.9, 0.3, 0.3), cube(10, center: true)),
    color((0.3, 0.7, 0.4), translate((15, 0, 0), sphere(6, fn:
32))),
  )),
  color: none,
)
```



Alpha is supported. Two equivalent forms:

```
color((0.9, 0.35, 0.35, 0.5), child) // RGBA in one 4-array
color((0.9, 0.35, 0.35), child, alpha: 0.5) // RGB + separate alpha
```

Interior geometry shows through translucent subtrees, so a coloured cover can reveal the shape's internal features.

The `scadypst()` compile call

```
scadypst(node, bin: (:), font: none, fn: 32)
```

- **node**: the tree returned by any DSL helper, usually a top-level union or difference, or a single primitive.
- **bin**: a dict of sidecar bytes referenced by `import-mesh(file)`. The keys match the string passed to `import-mesh`, and the values are `read("path", encoding: none)`.
- **font**: TTF or OTF bytes for `scad-text(...)`. One font per compile.
- **fn**: the default `$fn` for the whole compile. A per-primitive `fn` overrides it.

It returns the same PLY bytes that `compile-scad` returns, ready to pass to `render-ply`.

Name clashes with Typst built-ins

A `glob-import (#import "...": *)` shadows seven of Typst's own functions:

`scale`, `rotate`, `circle`, `square`, `ellipse`, `polygon`, `color`. (Additionally, the OpenSCAD text primitive is re-exported as `scad-text` to avoid shadowing `#text`.)

Two ways around it:

```
// Option 1: namespace the plugin
#import "@preview/maquette-scad:0.1.0"
#let part = maquette-scad.scadypst(
  maquette-scad.difference(
    maquette-scad.cube(20, center: true),
    maquette-scad.sphere(12, fn: 48),
  )
)

// Option 2: glob-import the plugin's `scad-*` aliases
#import "@preview/maquette-scad:0.1.0": scad-color, scad-scale, scad-rotate, scad-circle, scad-square,
scad-ellipse, scad-polygon
```

Both leave Typst's own `#text` / `#circle` / ... intact. The `scad-*` aliases live alongside the unprefix names, so you can mix approaches within one document.

Compiling .scad sources with compile-scad

For a standalone .scad file, one call is enough:

```
#render-scad(read("part.scad"))
```

Real-world .scad files rarely stand alone: they use `/ include` a library, import an STL or OBJ for a fastener, or ship a font for embossed text. The wasm sandbox has no filesystem, so every sidecar is passed explicitly as bytes:

```
#compile-scad(read("main.scad"),
  files: (
    "utils.scad":      read("utils.scad"),
    "gears.scad":      read("gears.scad"),
    "MCAD/nuts_and_bolts.scad": read("MCAD/nuts_and_bolts.scad"),
  ),
  bin: (
    "mount.stl":      read("mount.stl", encoding: none),
  ),
  font: read("logo.ttf", encoding: none),
  fn: 48,
  trace: "trace.log",
)
```

Options:

- **source** (positional string): the .scad text, almost always `read("path.scad")`.
- **files** (dict): resolves every use `<name>` and `include <name>` in the source. The keys are the exact names the .scad uses, and nested paths like `MCAD/foo.scad` are keys with slashes.
- **bin** (dict): sidecar mesh files referenced by `import()`. The keys match the `import` argument. STL and OBJ are recognised; DXF, 3MF and AMF are not.
- **font** (bytes): a TTF or OTF file used by `text()`. One font per compile.
- **fn** (int): the default `$fn` for this compile. Any per-primitive `fn:` overrides it.
- **trace** (string): dumps the evaluator trace to this path. It helps when a nested `for` or `module` is not producing what you expected.

Inspection: scadypst-info / compile-scad-info

Returns a Typst dict with the final geometry's stats without building the PLY. Useful for laying parts out by their real size, annotating a document with computed volumes, or failing early if a compile produced empty geometry.

Fields: `bbox_min`, `bbox_max`, `center`, `radius`, `volume`, `surface_area`, `num_tri`, `num_vert`, `genus`.

```
#let info = scadypst-info(cube(20, center: true))
#raw(block: true, lang: "yaml",
  "volume:      " + str(info.volume) + " mm³\n" +
  "surface_area: " + str(info.surface_area) + " mm²\n" +
  "num_tri:      " + str(info.num_tri) + "\n" +
  "bbox_min:     " + repr(info.bbox_min) + "\n" +
  "bbox_max:     " + repr(info.bbox_max)
)
```

```
volume:      8000 mm³
surface_area: 2400 mm²
num_tri:     12
bbox_min:    (-10, -10, -10)
bbox_max:    (10, 10, 10)
```

`compile-scad-info(src, files: (:), bin: (:), font: none, fn: 32)` is the .scad-source variant with the same return shape.

Decomposing: scadypst-parts / compile-scad-parts

Splits the final geometry into its connected components via `Manifold::decompose()` and returns each as its own PLY. The return type is `array<bytes>`; each element is renderable via `render-ply` in isolation.

Useful for laying out multiple pieces separately (laser-cut sheet layouts, exploded-view annotations, per-part BOM tables).

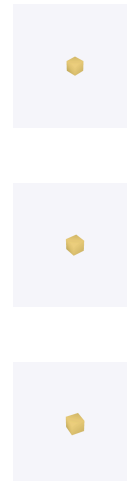
```
#let parts = scadypst-parts(union(
  translate((0, 0, 0), cube(4, center: true)),
```

Got 3 parts:

```

translate((10, 0, 0), cube(4, center: true)),
translate((20, 0, 0), cube(4, center: true)),
))
Got #parts.len() parts:
#for p in parts {
  show-part(p, width: 25%)
  h(0.3em)
}

```



`compile-scad-parts(src, files: (:), bin: (:), font: none, fn: 32)` is the `.scad-source` variant.

Vertex reduction: `simplify(child, epsilon)`

Collapses edges shorter than `epsilon`, measured in the input's units. It is a Douglas-Peucker style pass that strips micro-detail from booleans, or slims a mesh before you ship it through PLY or SVG.

It is dimension-agnostic: pass a 2D shape (`CrossSection`) or a 3D solid (`Manifold`), and the op dispatches on the child.

```

#let ring = union(..range(64).map(i => {
  let a = i * 360deg / 64
  translate((calc.cos(a) * 10, calc.sin(a) * 10),
    circle(1.2, fn: 8))
}))
#let ring-info = scadypst-info(ring)
#let simple-info = scadypst-info(simplify(ring, epsilon: 0.5))
#raw(block: true, lang: "yaml",
  "before -> num_tri: " + str(ring-info.num_tri) + "\n" +
  "after -> num_tri: " + str(simple-info.num_tri)
)

```

```

before -> num_tri: 1152
after -> num_tri: 112

```

`epsilon` is in model units: keep it tiny for a fine mesh, and make it generous when you want to knock a curve down to a coarse polyline.

Smooth shading: `calculate-normals(child, sharp_angle)`

Attaches per-vertex normals to a 3D solid so maquette can smooth-shade curved surfaces while keeping crisp edges. Adjacent faces whose dihedral angle exceeds `sharp_angle` (degrees) stay sharp, and the rest blend. Run it after the final booleans and hulls, since those discard any existing normals.

The effect only shows under smooth shading. `openscad-view` is flat by design, so switch shading: `"smooth"` (and drop `openscad-view`) for the render call.

```

#let part = calculate-normals(
  difference(sphere(10, fn: 24), cube(12, center: true)),
  sharp_angle: 30,
)
#render-ply(scadypst(part),
  width: 3.5cm, shading: "smooth",
  azimuth: 30, elevation: 25)

```



Lower `sharp_angle` = more edges classed as sharp (crisp box corners). Higher = more edges smoothed (softer curves).

Two ways to add smooth normals

`calculate-normals` (above) is a **tree op**: it smooths whatever subtree it wraps. Use it when only one region of a bigger model should render smooth, say the sphere but not the cube parts.

The end-call helpers `scadypst(..., smooth-normals: 30)` and `compile-scad(..., smooth-normals: 30)` are the **whole-scene** shortcut: they run the same `calculate_normals` pass on the final mesh right before it's serialized. Same output, less typing when you want the entire model smoothed:

```
// tree-op form: smooths just the sphere
scadypst(union(
  calculate-normals(sphere(10, fn: 24), sharp_angle: 30),
  translate((15, 0, 0), cube(6, center: true)),
))

// end-call form: smooths everything
scadypst(my-model, smooth-normals: 30)
compile-scad(read("part.scad"), smooth-normals: 30)
```

Either way, the render call needs a smooth shading mode (shading: "smooth") to actually consume the normals.

Ray casting: `scadypst-raycast` / `compile-scad-raycast`

Fires a segment from origin to end at the evaluated geometry and returns an array of hits sorted by distance:

```
(
  ( face_id: <int>, distance: <float>,
    position: (x, y, z), normal: (x, y, z) ),
  ..
)
```

Use it to sample a terrain height under an XY coordinate, pick the face at a screen click, or check line-of-sight between two points. Empty array means the segment missed.

```
#let part = union(
  cube(10, center: true),
  translate((0, 0, 12), sphere(5, fn: 32)),
)
#let hits = scadypst-raycast(part,
  (0, 0, 30), (0, 0, -30))
#raw(block: true, lang: "yaml",
  "hits: " + str(hits.len()) + "\n" +
  hits.map(h =>
    " d=" + str(calc.round(h.distance, digits: 2)) +
    " z=" + str(calc.round(h.position.at(2), digits: 2))
  ).join("\n")
)
```

```
hits: 4
d=0.22 z=17
d=0.38 z=7
d=0.42 z=5
d=0.58 z=-5
```

`compile-scad-raycast(src, origin, end, files: (:), bin: (:), font: none, fn: 32)` is the `.scad-source` variant.

Direct 2D → SVG: `scadypst-svg` / `compile-scad-svg`

For sources that resolve to a 2D shape (circle, square, polygon, text, hulls of 2D things, offset, slice, projection), the plugin can emit an SVG document directly, with no maquette in the loop and no rasterizer. The result is resolution-independent, Inkscape-editable, and laser-cutter-ready.

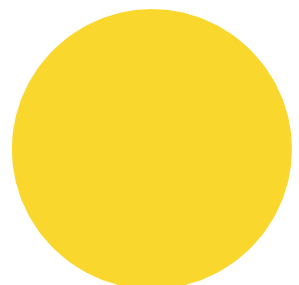
Errors if the tree resolves to a 3D solid.

```
#image(scadypst-svg(difference(
  union(
    circle(10, fn: 64),
    translate((14, 0), square(20, center: true)),
  ),
  circle(4, fn: 48),
)), width: 65%)
```



`slice(child, z: 0)` composes cleanly with `-svg`: 3D model → horizontal cross-section → vector contour, in one call.

```
#image(scadypst-svg(slice(sphere(10, fn: 64), z: 4)), width: 60%)
```



`compile-scad-svg(src, ...)` is the `.scad-source` variant.

Multi-file projects: `compile-scad-tree`

A project split across many files with use `<..>` and `include <..>` compiles in one call. `compile-scad-tree` reads the entry file and follows its include graph for you, so there is no file list to maintain:

```
#import "@preview/maquette-scad:0.1.0": openscad-view, render-scad-tree

#render-scad-tree(
  "Cyclone.scad",
  root: "cyclone-src/",
  read: p => read(p),
  ..openscad-view,
  azimuth: 35,
  elevation: 20,
  up: (0, 0, 1),
  background: none,
)
```

`root` is the source folder joined onto each discovered path, and `entry` is the top file inside it. Pass a `read: lambda` scoped to your document, `read: p => read(p)`, so the sources are read from **your** project; a package's own `read()` cannot reach your files. The whole Cyclone-PCB-Factory below, all 69 files, comes from that single call:

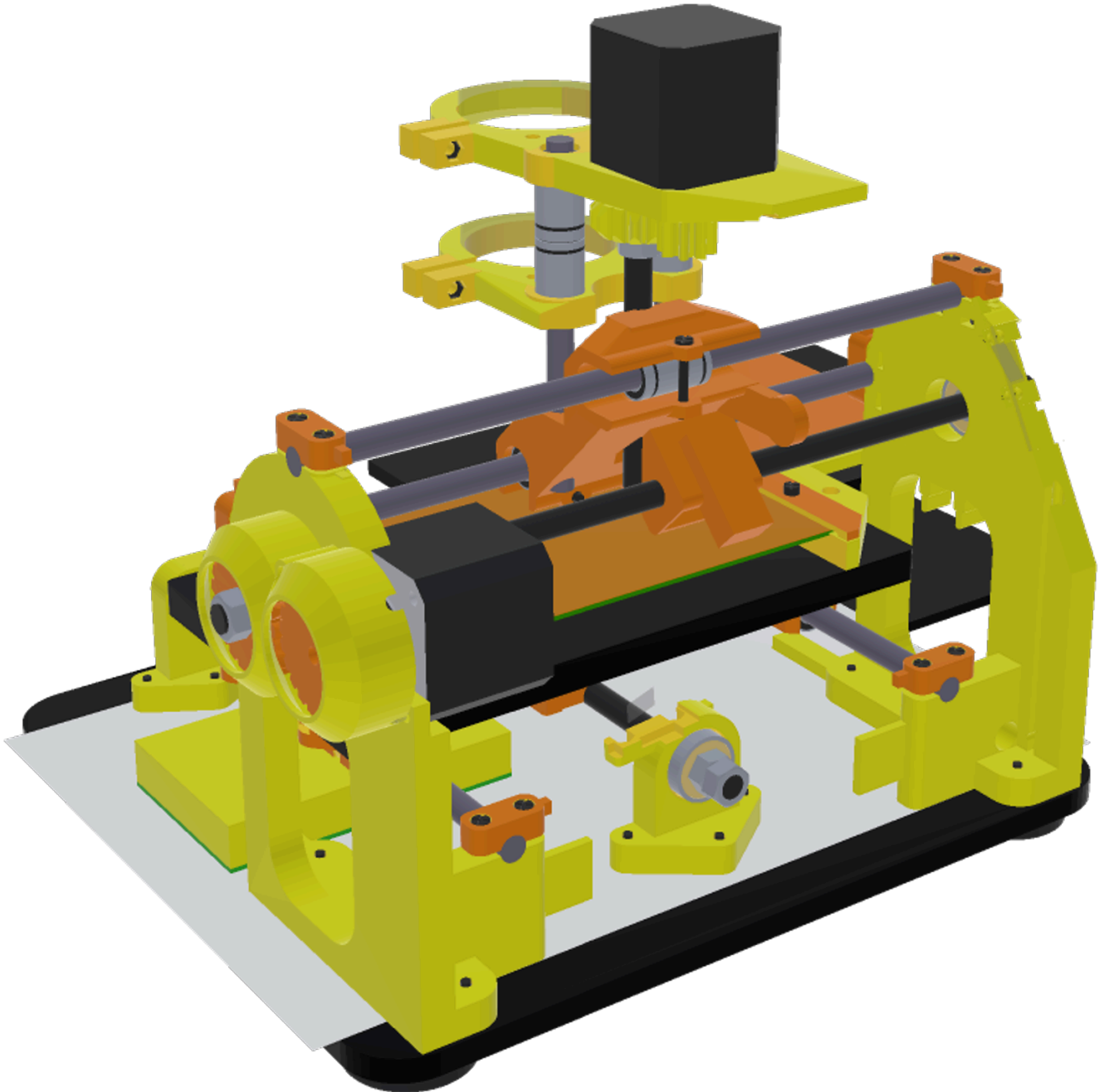


Figure 1: The full Cyclone-PCB-Factory from one `compile-scad-tree` call.

Need the files as a dict instead (to edit one before compiling)? `scad-collect(entry, root:)` returns `path → source` for `compile-scad(main, files: ...)`.

What **compile-scad** accepts

If your `.scad` source works in the OpenSCAD editor, it should work here; the evaluator covers everything a typical file uses. The known gaps are `surface()` (heightmap import), `import()` of DXF, the `$vpr` / `$vpt` / `$vpd` / `$vpf` viewport variables, and adaptive tessellation from `$fa` / `$fs` (we use `$fn` or the per-primitive `fn:` instead). Any external STL or OBJ mesh referenced by `import()` must be routed through the `bin:` argument, since the plugin has no filesystem access.

The exhaustive per-feature tracker lives at `crates/maquette-scad/FIDELITY.md`.

Two wasm modules cooperate. `maquette-scad.wasm` (this crate) compiles geometry to PLY. `maquette.wasm` renders it. They exchange nothing but the PLY blob, so you can also feed a `scadypst(...)` output into `maquette-gltf` (wrapped in a minimal `.gltf`) when you want PBR shading on procedural geometry.